

RESIDUAL STRENGTH AFTER TRANSVERSE IMPACT IN COMPOSITES

Part: **1.4 – COMPOSITE MATERIALS**

Chapter: **1421**

Title: **RESIDUAL STRENGTH AFTER
TRANSVERSE IMPACT IN
COMPOSITES**

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1 INTRODUCTION

1.1 OBJECT

The aim of this chapter is to present approximation models for residual strengths after BVID, capable to capture the main sensitivities of this failure mode to laminate definition parameters and damage severity, with a reasonable level of accuracy, and still maintaining a significant level of simplicity, enabling their practical use in structural sizing.

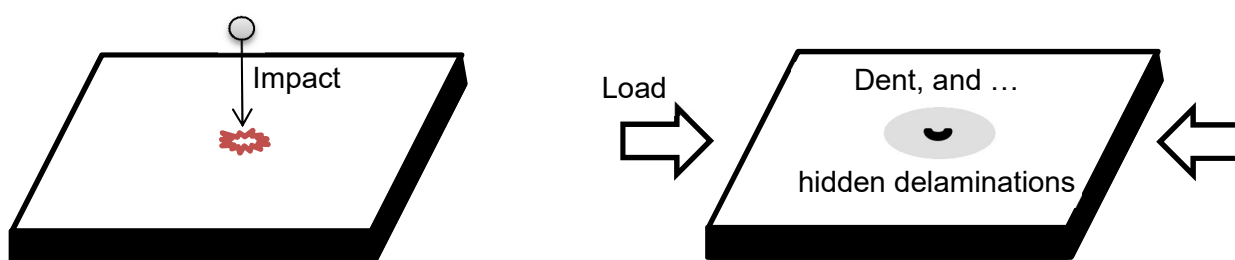


Figure 1. Sketch of the CAI problem

Reductions of design variables and other simplifications have been used extensively. Curve-fitting of tests data is required to adjust the resulting material model. Among other consequences, this limits the applicability of current proposal to conventional designs.

Failure criterion to perform the check strength is a modified version of the maximum strain criterion (ignoring shear strain in plies, see chapter 3405 “RESIDUAL STRENGTH AFTER BVID IMPACT”). That criterion is considered applicable for laminates following the conventional design rules. That criterion compares gross/pristine axial strain in plies, at fibre directions, calculated using classical laminates theory, with the allowable strains (per direction) provided by the method described in this document.

1.2 SCOPE AND LIMITATIONS

Method applicable to:

- ✓ Fibrous composite solid laminates (made of UD-tape or biaxial-fabrics)
- ✓ Stacking sequences following the typical design rules
- ✓ Transverse low velocity impacts causing BVID damages
- ✓ Supported plates/shells (locations at certain distance of edges or stiffeners)

Limitations:

- ✓ Effect of distance to stiffeners not captured by this model. Nevertheless, actual proposal is considered conservative, as energy dissipation by means of elastic bending will mitigate damage severity in a wider panel (compared with the width of standard CAI specimens)
- ✓ Transverse impacts near plate/hole edges, and edge impacts, are not covered by the model presented in this chapter

2 METHOD DISCUSSION

2.1 Generalities

2.1.1 REGULATORY ASPECTS

Airworthiness regulations dedicated to composite airframes, in particular those affecting primary structures, request that their residual strength be capable to carry ultimate load during the whole Design Life Goal with some level of damages present within the materials, including allowable manufacturing defects and barely visible impact damages. See further explanations in Ref. 1 §1.4, or regulators documents EASA's AMC 20-29 or FAA's AC No: 20-107B.

2.1.2 BUILDING THE APPROXIMATION MODEL

As explained in the introduction, the aim of this chapter is to present approximation models for residual strengths after BVID, capable to capture the main sensitivities of this failure mode to laminate definition parameters and damage severity, with a reasonable level of accuracy, and still maintaining a significant level of simplicity, enabling their practical use in structural sizing.

The main tools used to define the approximation model presented in this document were:

- ✓ Function approximation by the tensor product bases method
- ✓ Correlation and regression analysis of tests data
- ✓ Dimensional analysis

The method finally presented has been checked, prior to release of this document, with several materials (see list of references), obtaining reasonable degrees of correlation with available tests data.

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2.1.3 NORMALIZATION RULES

Thickness

Residual tension and compression strengths obtained from test shall be normalized (to ultimate apparent stresses) using the nominal/theoretical thickness (*).

(*) It shall be checked that measured thickness values in coupons are within acceptable tolerances by company standards.

Stiffness (elastic moduli)

Residual tension and compression ultimate apparent stresses shall be converted into allowable strains using the theoretical apparent elastic modulus of the laminate. This modulus is calculated from material parameters defining ply stiffness (CPT , E_1 , E_2 , G_{12} , ν_{12}) and classical laminate analysis.

Future preferred strategy for normalization, regarding material ply stiffness, is to define one fixed set applicable to all ambient service conditions (cold to hot). Even E_1 and ν_{12} are normally naturally not significantly affected by temperature and wet-ageing, values of G_{12} , and E_2 in UD-tape, are affected by these changes. However, the macroscopic impact of these changes in laminate stiffness is typically low.

Statistics

Approximation model, adjusted by regression analysis of tests data, shall represent the mean trend. The B-basis correction shall be applied externally to the result of the regression models, assumed non-dependent (uncoupled) of the rest of design variables.

$$Strength = K_B \cdot (Strength)_{Mean}$$

Environment conditions

Environment conditions are assumed non-dependent (uncoupled) of the rest of design variables and accounted by means of a knock down factor ($EKDF$) applied externally to the result of the regression models.

$$Strength = EKDF \cdot (Strength)_{RT/AR}$$

Additional details on these topics in chapters 1480 "DERIVATION OF DESIGN DATA IN COMPOSITES" and 1481 "DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES".

2.2 Dent Depth

The dent depth caused by an impact on a laminate is the selected indicator of detectability that will serve to define the BVID threshold. Thus, relating this magnitude with the parameter used in this

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model for measuring the damage severity shall be necessary. Damage severity is considered in this model by means of the parameter E/t^2 , i.e. the ratio between the impact energy and the squared thickness (*).

Dent depth relaxation

BVID threshold is set by either an energy cut-off (derived for each part by a damage threats analysis, being 35J a typical value), or by the visibility of the dent caused by the impact, whichever leads to the minimum damage severity. Reference values of detectable dents used in Airbus, for each type of inspection, are shown in following table.

Type of inspection	Detectable dent depth	
	after relaxation	after impact
DET	d = 0.3 mm	d ₀ = 1.0 mm
GVI	d = 1.3 mm	(*)

(*) The value of the dent depth after impact which does not relax beyond the threshold defined for the General Visual Inspection (GVI) is a material dependent variable, typically greater than 2mm. A typical value of 2.5mm is generally applicable (if no perforation). Nevertheless, using a test validated value for the specific material is strongly recommended.

Additional info about damage tolerance considerations in composite materials can be found in chapter 4910 "FATIGUE ASSESSMENT OF COMPOSITE PARTS".

Dent depth and impact severity

Available tests observations noted that the dent depth has not much sensitivity to laminate stacking sequence (or to laminate "directionality"). The relationship between the dent depth after impact (d_0), the laminate thickness (t) and impact energy (E) will be approximated with the following simple function:

$$E/t^\alpha = A \cdot (1 - e^{-B \cdot d_0/t}) \quad (*)$$

Coefficients A and B shall be obtained from a regression analysis of tests data, which should cover a wide range of severities (from barely visible dents to full penetration). Default value for exponent α is 2.0, but this parameter could be left open in the regression fitting (from the limited previous experience typical solutions for α are expected within the range 1.5 - 2.0).

Note that, due to the reduction of the dent depth with time and humidity, the detectable dent depths are defined after relaxation (d), whereas the expression above is expressed in terms of the dent depths measured after impact (d_0). The conversion may be different for each material (even some recommendations were given above).

(*) Under consideration the use of $E/(G_{IIC} \cdot t^2)$ to complete the normalization with a relevant material property, instead of just using E/t^2 , in order to compute the damage severity in this model using a non-dimensional parameter (pending further analyses of the available tests data).

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2.3 Damage Area

Reserved for future use.

Remark

Even a mathematical expression of the relationship between the laminate residual strength and the projected delaminated area caused by a low velocity impact is not strictly necessary for sizing the parts (detectability is controlled by the dent depth and damage severity can be monitored by the impact energy), providing an approximation model for this relationship shall be convenient for...

- ✓ Validating the threshold of reportable indications in quality inspections (NDI)
- ✓ Permitting the dispatch of the part including minor defects/damages (as single delaminations), detected in quality or maintenance inspections without repair.

2.4 Tension After Impact (TAI)

Since Tension After Impact (*TAI*) is usually not a key design driver, and analysis of available test data for *TAI* has not shown, compared with *CAI*, too much sensitivity to laminate configuration parameters, the proposed material model for this parameter has been simplified at maximum.

TAI residual strength, in terms of strain, will be considered a material constant for any laminate (following the standard design rules). Its design value shall be expressed in terms of allowable strain, as mean-value, at room temperature, plus the corresponding penalization factors for ...

- ✓ B-basis correction (typically K_B equal to 0.89)
- ✓ Hot ambient in wet-aged materials (typically $EKDF$ equal to 1.0 up to 70°W conditions)

In absence of dedicated tests data, as suggested per Ref. 5, Tension After Impact allowable strength can be conservatively set as:

$$TAI = \varepsilon_t / 3$$

...where ε_t is the tension allowable strain of the material, in fibre direction, tailored using not only F_{1t} (X_T) datum, but also available plain strength tests results data on multi-angular laminates.

A better alternative is usually the following:

$$TAI = 0.8 \cdot OHT_{qi} / E_{qi}$$

... where OHT_{qi} is the open hole tension strength (ultimate stress) of the quasi-isotropic material, property typically obtained during material qualification, and E_{qi} the nominal elastic modulus of that laminate.

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2.5 Bending After Impact (BAI)

In absence of dedicated test data, which will be the most frequent situation for any new CFRP material, previous tests experience has shown that *TAI* is a reasonable and conservative approximation for *BAI*.

Thus, the design value for *BAI* shall be made equal to *TAI*.

2.6 Shear After Impact (SAI)

Composite laminates covered by this methodology are expected to follow typical design rules, among others, built with a material stacking sequence capable to sustain a general set of loads just with the reinforcement fibres in tension or compression. Laminates with plies arranged at 0°, ±45° and 90° angles, with a minimum weight of plies of 10% per direction, homogeneously distributed, have that capability.

Provided that it is true, no design value for a Shear After Impact allowable strength (*SAI*) is required. Actually, *SAI* is not included in proposed failure criterion.

2.7 Compression After Impact (CAI)

The proposed model consists of a sum, or product, of two terms: one to account the effect for laminate directionality (fibres weight aligned with load) and the other to account the effect of the impact severity.

$$CAI = CAI_{\infty} + \Delta CAI_E \quad (\text{forms } CDTM-T04, CDTM-T05, CDTM-T06)$$

Or...

$$CAI = CAI_{\infty} \cdot K_E \quad (\text{form } CDTM-T07)$$

CAI after full penetration

The first term CAI_{∞} represents the residual strength of the laminate if impactor perforated the whole thickness. The sensitivity to laminate directionality is expected to be captured by the ratio E_x/E_{QI} (apparent elastic modulus ratioed with that of the QI laminate made of the same material) or by the stress concentration factor due to a circular open hole in an infinite plate (see chapter 3404 "STRENGTH IN NOTCHED LAMINATES"):

$$K_{T0H} = 1 + \sqrt{2 \left(\sqrt{\frac{E_x}{E_y}} - \nu_{xy} \right) + \frac{E_x}{G_{xy}}}$$

Pending on the directionality factor chosen, and on other circumstances, some regression models have been proposed recently within Airbus DS (see references 5 to 11) to approximate the delta strength over the reference CAI after penetration of the QI laminate, as:

$$\checkmark \quad CAI_{\infty} = CAI_{\infty QI} / (E_x/E_{QI})^C \quad (\text{CDTM-T04})$$

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$$\checkmark \quad CAI_{\infty} = CAI_{\infty QI} \cdot (3 / K_{TOH})^C \quad (CDTM-T05, CDTM-T07)$$

$$\checkmark \quad CAI_{\infty} = CAI_{\infty QI} \cdot [1 - C \cdot (E_x/E_{qi} - 1)] \quad (CDTM-T06)$$

The material parameters in the curve finally selected shall be obtained from a regression analysis of tests data, which should cover a wide range of stacking styles.

From the list of approximation functions shown above, the second one (labelled as “*CDTM-T05*”) is today the preferred for convenience, as the parameter measuring the laminate directionality K_{TOH} has a familiar meaning to stress analysts (3.0 corresponds to isotropic materials) and can be computed easily from the apparent moduli of the laminate in consideration.

This term is assumed to be uncoupled with the laminate thickness or with the BVID energy.

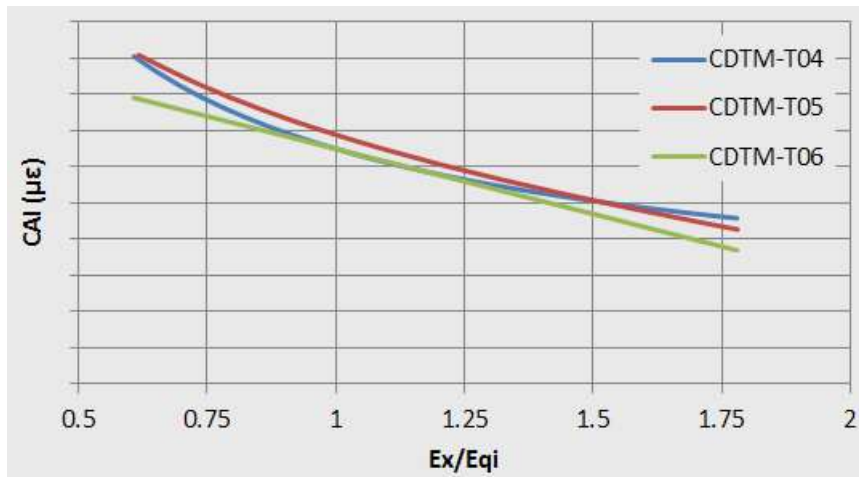


Figure 2. Trend of CAI_{∞} with laminate directionality using these models

The first of the abovementioned variants of this approximation (that labelled as *CDTM-T04*) corresponds literally to the proposal made in Ref. 5 for the Hexcel IMA/M21E CFRP material after revision and simplification of the conclusions resulted of its tests results analysis. The following variants were suggested in Ref. 6, etc..., with slightly different formats, for other materials with less abundance of tests results data.

CAI and impact severity

The second term ΔCAI_E represents the increase of residual strength if BVID energy was not capable to perforate the laminate. Damage severity is measured with impact energy and thickness using the parameter E/t^2 (see note * in page 4).

$$\Delta CAI_E = A / e^{B \cdot (E/t^2)^\alpha} \quad (\text{forms } CDTM-T04, CDTM-T05, CDTM-T06)$$

The alternative formula for this term, used with the form of CAI expressed as a product of functions (instead of a sum), is:

$$K_E = 1 + A / e^{B \cdot (E/t^2)^\alpha} \quad (\text{form } CDTM-T07)$$

Coefficients A and B shall be obtained from a regression analysis of tests data, which should cover a wide range of severities (from barely visible dents to full penetration). As for the dent depth equation,

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the recommended default value for α is 2.0, but this parameter could be left open for the search in the regression fitting analysis.

This term is assumed to be uncoupled with the laminate directionality.

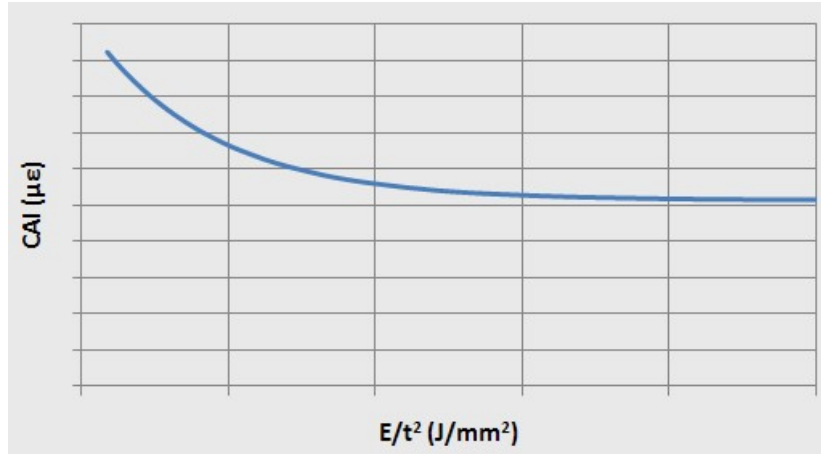


Figure 3. Trend of CAI strength with impact severity using this model

Final adjustment

Deviations from test results data in this model can't be justified exclusively by the natural material dispersion. Note simplifications reach the point of characterizing a laminate by only thickness and elastic modulus in load direction. Derivation of CAI allowable will not then be possible by just applying the B-basis correction ratio (typically K_B not less than 0.83 in fibrous composites laminates) to the regression curve. The mean trend represented by this approximation model shall normally need an adjustment to assure conservatism of the strength prediction. This shall be done by tweaking the parameter $CAI_{\infty QI}$, or by obtaining the best-fit adjustment parameters of the model with numerical optimization methods and adequate penalty functions.

It is important that the tests configurations provided to set-up this model cover sufficiently the design space, including laminates from low to medium thicknesses and specimens with few fibre proportion in load direction (0° plies $\approx 10\%$), quasi-isotropic variants, and up to directed laminates (0° plies $\geq 40\%$).

Approximation of CAI without tests data

In absence of dedicate tests data to properly adjust the models described above, the suggested approximation for CAI is to use the Open-Hole analysis, i.e. assume that failure will occur when the tangent strain in a fictitious hole in the laminate is equal to the allowable compression strain of the pristine material (see discussion and justification in Ref. 5). Note that no relaxations as the "characteristic distance" approach (popular in literature for open-hole failure predictions) shall be applied to the hole edge strain in this method. Thus, for a certain laminate at a particular direction of interest, the CAI strength in terms of allowable strain, would be approximated by:

$$CAI = \epsilon_{call} / K_{TOH}$$

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3 CONCLUSION: MATERIAL MODEL FOR TAI & CAI

3.1 Dent depth and impact energy

The impact energy level required to derive the residual strengths after BVID shall be the minimum between the applicable cut-off ($E_{cut-off}$) and the value obtained from this expression:

$$E/t^\alpha = A \cdot (1 - e^{-B \cdot d_0/t})$$

...where:

- ✓ E is the impact energy, in J
- ✓ t is the thickness of the laminate, in mm
- ✓ d_0 is the detectable dent depth after impact, and shall be made 1mm, if inspection program for the part includes detailed inspections (DET), or 2.5mm (TBC) if only includes general visual inspections (GVI).
- ✓ A and B are parameters of the model to be adjusted from tests data

Then, the impact energy at the BVID threshold can be expressed as:

$$E_{BVID} = \min [E_{cut-off}, t^\alpha \cdot A \cdot (1 - e^{-B \cdot d_0/t})]$$

3.2 TAI

The Tension After Impact residual strength (TAI) of a laminate shall be obtained as:

$$TAI = K_B \cdot EKDF \cdot TAI_\infty$$

...where:

- ✓ K_B is the B-basis correction factor
- ✓ $EKDF$ is the applicable Environmental Knock Down Factor
- ✓ TAI_∞ is the TAI residual strength after penetration, in $\mu\epsilon$

3.3 CAI

The Compression After Impact residual strength (CAI) of a laminate, for each of the relevant material directions 0° , $\pm 45^\circ$ and 90° shall be obtained as:

$$CAI_\infty = \begin{cases} K_B \cdot EKDF \cdot [CAI_{\infty QI} / (E_x/E_{qi})^C + A / e^{B \cdot (E/t^\alpha)}] & (CDTM-T04) \\ K_B \cdot EKDF \cdot [CAI_{\infty QI} \cdot (3 / K_{TOH})^C + A / e^{B \cdot (E/t^\alpha)}] & (CDTM-T05) \\ K_B \cdot EKDF \cdot \{ CAI_{\infty QI} \cdot [1 - C \cdot (E_x/E_{qi} - 1)] + A / e^{B \cdot (E/t^\alpha)} \} & (CDTM-T06) \\ K_B \cdot EKDF \cdot \{ CAI_{\infty QI} \cdot (3 / K_{TOH})^C \cdot [1 + A / e^{B \cdot (E/t^\alpha)}] \} & (CDTM-T07) \end{cases}$$

...where:

- ✓ K_B is the B-basis correction factor

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- ✓ E_{KDF} is the applicable Environmental Knock Down Factor
- ✓ $CAI_{\infty QI}$ is the CAI residual strength after penetration of the QI laminate, in $\mu\epsilon$
- ✓ E_X is the elastic modulus of the laminate in load direction, in MPa
- ✓ E_{QI} is the elastic modulus of the QI laminate of this material, in MPa
- ✓ K_{TOH} is the stress concentration factor in the laminate due to an open circular hole
- ✓ E is the impact energy, in J
- ✓ t is the thickness of the laminate, in mm
- ✓ A, B , and C are parameters of the model to be adjusted from tests data (optionally also exponent α)

Last form *CDTM-T07* shall be typically the preferred for new materials or revision of design datasets of legacy materials.

Example of CAI calculation for a dummy laminate:

a) Required inputs

Detectability and impact damage threat considerations

Impact cut-off energy: $E_{CO} = 35 \text{ J}$

Detectable dent depth after impact: $d_0 = 2.0 \text{ mm}$

Material properties

Ply thickness: $t_{ply} = 0.2 \text{ mm}$

Ply moduli: $E_1 = 150000 \text{ MPa}$, $E_2 = 8500 \text{ MPa}$, $G_{12} = 4200 \text{ MPa}$, $\nu_{12} = 0.3$

Damage model: $E_{BVID} = \min [35J, t^2 \cdot 6.0 \cdot (1 - 1/e^{4.5 \cdot d_0/t})]$

CAI model: $CAI = 0.83 \cdot 0.90 \cdot [3000 \cdot (3 / K_{TOH})^{0.5} + 5000 / e^{1.5 \cdot (E/t^2)}]$

Laminate stacking (45/-45/0/0/90/0/0/-45/45/0)S (thickness $t = 4.0 \text{ mm}$)

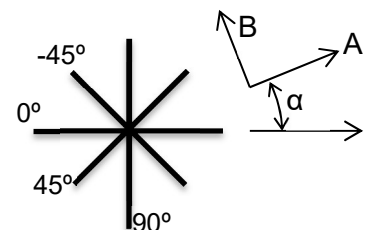
b) Estimation of CAI

Laminate thickness: $t = 4.0 \text{ mm}$ (20 plies)

BVID impact energy: $\min [35, 4.0^2 \cdot 6.0 \cdot (1 - 1/e^{4.5 \cdot 2.0/4.0})] J = \min [35, 86] J = 35 J$

CAI values for each ply orientation:

α (°)	E_α (MPa)	E_β (MPa)	$G_{\alpha\beta}$ (MPa)	$\nu_{\alpha\beta}$	K_{TOH}	CAI ($\mu\epsilon$)
0°	87271	34656	17939	0.4350	3.677	2046
±45°	46463	46463	24808	0.2951	2.812	2337
90°	34656	87271	17939	0.1728	2.687	2390



4 APPENDICES

FUTURE WORK

Still to be done:

- ✓ Approximation model of damage area. The dispersion of measured projected damages areas is typically big. Instead of regressions, as done for other magnitudes, lines limiting lower and upper bounds might be preferable (each limit to be used in different contexts)
- ✓ Approach/sizing-policy for residual strength after transverse impact damage near edges
- ✓ Develop tool to automate calibration of approximation model from tests data
- ✓ Develop capabilities to virtualize *TAI* and *CAI* tests by advanced numerical simulation
- ✓ Compare approximation models for different materials completing the normalizations to non-dimensional / non-material-dependent variables, e.g. $E/t^2 \rightarrow E/(G_{IIC}t^2)$, $CAI \rightarrow CAI/\epsilon_c$, and investigate further generalizations

VALIDATION CASES

This approach has been used to set up approximation models for several materials, from tests data of different exhaustiveness and range coverage (references 5 to 11).

The most exhaustive *TAI* & *CAI* coupons tests program ever done at Airbus DS was performed with the Hexcel IMA/M21E UD-Tape material proposed for the Aeron-AS2 project. Successive analyses of these test results data (Ref. 4 and Ref. 5) inspired the model finally presented.

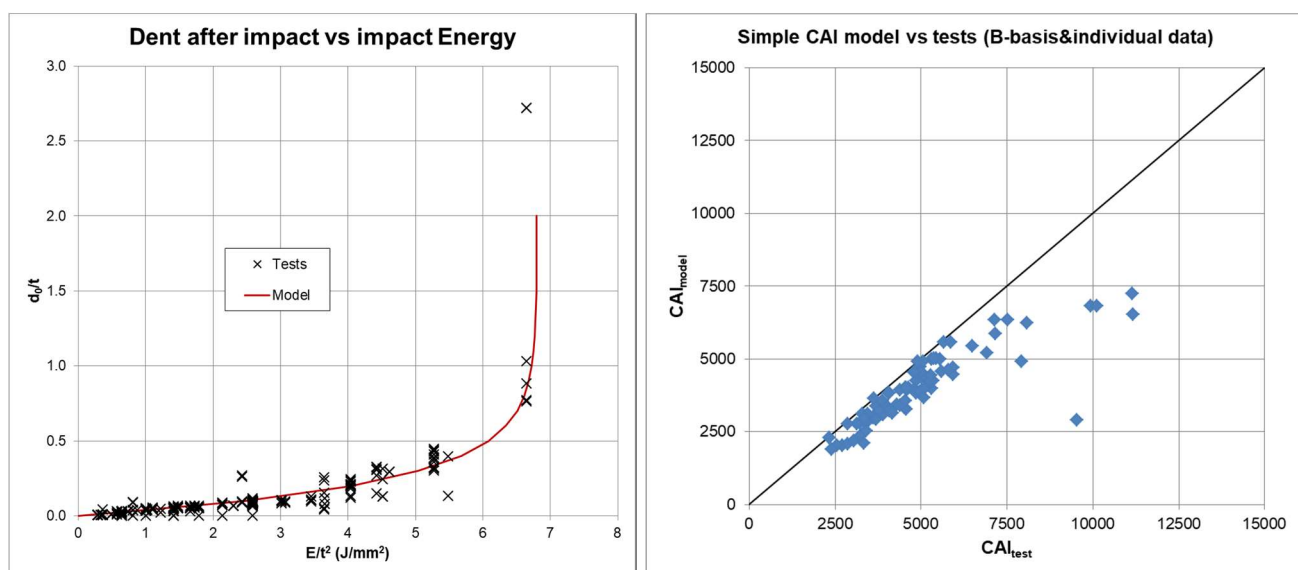


Figure 4. Approximation mode: dent depth and CAI for IMA/M21E (from Ref. 5)

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Remarks:

- ✓ Note dispersion of the dent depths, one of the damage measurable indicators, is high. Dispersions of observed/projected delaminated areas were even higher. This motivated not using any of this parameters as the variable to control the damage severity (E/t^2 instead).
- ✓ Model for CAI is overly conservative for greater laminate thicknesses (CAI values over 7500µε in this case), where the BVID cut-off results in very low damage severities

TEMPLATES FOR PREPARATION OF TESTS RESULTS DATA

In order to derive material design values, it is strongly recommendable to prepare results data in the format proposed by following table. Indeed it is suggested to directly request it to the laboratory in charge of doing the tests.

Laminate					Impact	Damage			Condition		Strength
Coupon ID	%0°	%45°	%90°	t (mm)	E (J)	A (mm ²)	d ₀ (mm)	d (mm)	T (°)	Age-ing	CAI (*) (MPa)
#1234	16.67	66.67	16.67	4.345	35	1200	2.45	1.87	20	Dry	340.5

Table 1. Presentation of individual CAI (or TAI) tests results before data analysis

MINIMUM RECOMMENDED TESTS MATRIX

Provided the rest of standard characterization tests of a fibrous composite materials area also available (inc. multi-angular plain tension and plain compression tests), for setting up the damage tolerance material model the following minimum tests matrix is proposed (from Ref. 2).

Properties and type of specimens		Test method	(10/80/10)		(25/50/25)		(50/40/10) UD or (40/20/40) Fab.	
			RT/Dry	H/W	RT/Dry	H/W	RT/Dry	H/W
TAI	1mm < t < 3mm	TEA-T-SP-170011 (penetration impact)			X	X		
CAI	t~4mm	AITM1-0010 impacts acc. to TEA-T-SP-170011			X	X		
	t~3mm	TEA-T-SP-170011	X	X	X	X	X	

Table 2. Minimum test matrix for setting up the TAI & CAI model

Typically the project promoting the screening or qualification of the new material shall add to that list one or more laminates representative of the part design.

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ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
AR	-	As Received material (no moisture-aged). Within this report, this condition has been considered equivalent to the “Dry” material condition.
BAI	$\mu\epsilon$	Bending After Impact residual strength of laminate including a BVID
BVID	-	Barely Visible Impact Damage, either determined by the visibility threshold (typ. dent depth after relaxation not less than 1.3 mm for general visual inspections or 0.3 for detailed inspections), or by the impact energy cut-off (typ. value of 35 J, function of A/C component location).
CAI	MPa , $\mu\epsilon$	Residual Compression strength After transverse Impact in a fibrous composite laminate including a BVID
CFRP	-	Carbon Fibre Reinforced Polymer/Plastic
CLT	-	Classical Lamination Theory
CTP	mm	Cured Ply Thickness (thickness of the composite material ply)
d	mm	Dent depth after relaxation
d_0	mm	Dent depth after impact
DET/DVI	-	Detailed Visual Inspection
DT	-	Damage Tolerance
DV	-	Design Value
E	MPa , J	Elastic (Young's) Modulus or impact energy, from context
EKDF	-	Environmental knock down factor, i.e. the correction factor to apply to a property obtained at room temperature in order to get the value at applicable hot/wet conditions
F_{1t}	MPa	Tension strength of FRP material in warp direction
F_{1c}	MPa	Compression strength of FRP material in warp direction
FRP	-	Fibre Reinforced Polymer/Plastic
G	MPa	Shear elastic modulus
G_{IIC}	J/m ²	Energy-release-rate / fracture-toughness in mode II (sliding / interlaminar shear) of the material. Remark: units should be consistent with magnitudes in expression $E/(G_{IIC} \cdot t^2)$, thus E in J, G_{IIC} in J/m ² and t in m match.
GVI	-	General Visual Inspection
K_B	-	B-basis knock down factor (ratio B-basis to mean)
K_{TOH}	-	Stress concentration factor of a circular open hole in an infinite composite plate subjected to axial in-plane load
PC	-	Plain Compression in-plane test done in a fibrous composite laminate
PT	-	Plain Tension in-plane test done in a fibrous composite laminate
OHT	MPa	Ultimate open hole tension strength
QI	-	Quasi-Isotropic laminate (typically with 25% plies at 0°, 25% at 90°, 25% at 45°, and 25% at -45°)

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QTR	-	Qualification Tests Report
RT	-	Room Temperature ambient conditions
SAI	$\mu\epsilon$	Shear After Impact residual strength of laminate including a BVID
t	mm	Thickness of the laminate
TAI	MPa , $\mu\epsilon$	Residual Tension strength After transverse Impact in a fibrous composite laminate including a BVID
TBC	-	To be completed or confirmed
UD	-	Uni-directional
x	-	As subscript, refers to the laminate X-direction, typically aligned with the fibre/warp direction of the plies at 0°, and with the applied load in tension/compression tests
y	-	As subscript, refers to the laminate Y-direction (normal to X-direction in-plane)
ϵ_{1t}	$\mu\epsilon$	Tension strength of the material in direction, in terms of allowable strain
ϵ_{1c}	$\mu\epsilon$	Compression strength of the material in warp direction, in terms of allowable strain
ϵ_{call}	$\mu\epsilon$	Allowable compression strain of the material, typically equal to ϵ_{1c}
ν	-	Poisson's ratio

REFERENCES

	Document Reference	Issue	Title
Ref. 1	ME-T-NT-130045	A	FAILURE CRITERION: RESIDUAL STRENGTH AFTER BVID IMPACT IN COMPOSITE PARTS (DAMAGE TOLERANCE)
Ref. 2	TAE-T-MM-170009	C	GUIDELINES FOR RESULTS ANALYSIS OF COUPONS TESTS IN COMPOSITE MATERIALS
Ref. 3	TEA-T-SP-170011	A	TEST METHOD FOR DETERMINATION OF THE STRENGTH AFTER IMPACT OF SOLID LAMINATES MADE OF FIBRE REINFORCED PLASTICS.
Ref. 4	TAE-AE-NT-160002	A	AERION AS2 WING LAMINATES TAI AND CAI TESTS RESULTS ANALYSIS
Ref. 5	TEA-T-NT-180036	C	LAMINATES STRENGTH. DISCUSSION ON TESTS RESULTS FOR IMA/M21E: PLAIN STRENGTH AND RESIDUAL STRENGTH AFTER IMPACT.
Ref. 6	TEA-T-NT-180037	C	RESIDUAL STRENGTH AFTER IMPACT OF CFRP HEXCEL AS4/8552 UD-TAPE LAMINATES

RESIDUAL STRENGTH AFTER TRANSVERSE IMPACT IN COMPOSITES

Ref. 7 TEA-CS-NT-180110

B RESIDUAL STRENGTH AFTER IMPACT OF
CFRP HEXCEL HITAPE/RTM6 UD-TAPE
LAMINATES

Ref. 8 TEA-CS-NT-180111

A RESIDUAL STRENGTH AFTER IMPACT OF
CFRP CYTEC/SOLVAY AS4/APC-2 (ISC) UD-
TAPE LAMINATES

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 3. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub- chapters affected</i>
1.0 04/08/2020	G. Baños	First issue	All pages
1.1 21/01/2022	G. Baños	Title modified, alternative approximation for TAI, and alternative form for CAI as product (not sum) of univariate functions	Pages 6, 7, 9, 10, 12-14
1.2 04/07/2022	G. Baños	Exponent of the thickness at the impact severity parameter (E/t^2) optionally left open in the search of the regression fitting analysis	Pages 4, 7 - 11
1.3 01/12/2022	G. Baños	Added conservative estimation of CAI in absence of dedicated tests data to adjust the proposed models	Pages 8, 14

Table 4. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños 01/12/2022	Florian Glock 09/12/2022

Table 5. Approval signatures table for last revision of this chapter